

ACTIVITY 21 — LEARNER WORKSHEET**Root Cause Analysis with 5 Whys, Fishbone and Pareto****TYPE**
Workshop**DURATION**
60 minutes**MAPS TO**
LO3**TOPIC**
5 - Monitor and Control the
Project**What you are doing**

Contoso Learning Pte Ltd is a training provider with 14,000 active learners. Its course registration portal is 9 years old. Registration takes a median of 11 minutes and 34% of attempts are abandoned. Learner communications consume 24 staff-hours per week across 3 administrators. 62% of learner traffic is mobile, against a portal that is not responsive. Legacy consent screens do not meet current PDPA requirements. The project modernises course registration and learner communications.

What you will produce: pareto analysis, fishbone, 5whys — submitted to artifacts/.

What you need before you start

- Lab 20 CPI decline and its systemic signal
- Lab 08 definition of done
- Lab 03 issue path

Your steps

1. Take the defect data. This is the real defect log for the Contoso Training Portal Upgrade, covering sprints 1 to 7 - the same period over which Lab 20 showed CPI declining from 1.022 to 0.919. Every defect here was found after the developer marked the story done.
2. Build the Pareto chart. Enter these category/count pairs exactly:
3. Re-run Pareto by cost, and notice the disagreement. Frequency is not impact. Re-rank the same data by rework hours.
4. Fishbone the top category. Set the problem statement (the fish head) to exactly:
5. Drive to root cause with 5 Whys. Then work the chain. Each "why" must be answerable with evidence, not opinion.
6. Verify the root cause before acting. A root cause you have not tested is a hypothesis. Apply the two standard tests.
7. Design corrective and preventive actions. The distinction is exam-relevant and practically important.
8. Answer the exam-style scenarios

Your working space

Complete every field. Keep this worksheet — it is your reference for the rest of the course.

Step 1 — Take the defect data**Step 2 — Build the Pareto chart**

Step 3 — Re-run Pareto by cost, and notice the disagreement

Step 4 — Fishbone the top category

Step 5 — Drive to root cause with 5 Whys

Step 6 — Verify the root cause before acting

Step 7 — Design corrective and preventive actions

Step 8 — Answer the exam-style scenarios

Deliverable checklist

- 21-pareto-analysis.md - the Pareto table by frequency with cumulative percentages, the vital few identified at the 80% cut, plus the second Pareto by rework hours and a written statement of what changed between them and why.
- A screenshot or export of your Pareto Chart tool output.
- 21-fishbone.md - all six categories populated with at least two causes each, the branches ranked by likely contribution with supporting evidence, and your chosen branch identified.
- A screenshot or export of your Fishbone tool output.
- 21-5whys.md - the five-why chain with evidence at every level, the stated root cause, and the results of the therefore test and the removal test.
- The end-to-end causal chain linking the root cause to the Lab 20 CPI decline.

- 21-corrective-actions.md - at least 8 actions, each classified as corrective, preventive or defect repair, with owner, date and success measure.
- Written answers to the three scenarios.